1. Simple Stresses and Strains

1.1 Introduction

Whenever a load is applied to a body, its shape changes i.e. the body gets deformed and the process is called deformation. The amount by which body is deformed depends upon the amount of load and the nature as well as the geometrical properties. It has been experimentally found that the cohesive forces between molecules of a body resist the deformation and force of resistance increases with increment in deformation. It has also been observed that the process of deformation stops when the force of resistance is equal to the external load. If the force of resistance is less then the external load, the material gets failed.

1.2 Stress: When an external force or load is applied on a body internal resistance is set up. Let us consider a bar acted upon by a Force P along its axis as in Fig 1.1(a). If the bar AC is cut in two sections at B, it results in Fig 1.1(b). The externally applied forces to one side of the cut part must be balanced by internal forces developed at the cut.

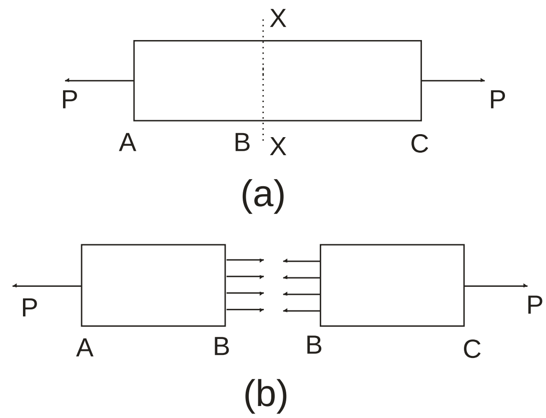


Fig. 1.1

If the internal resistance offered by the section against the deformation be assumed to be uniform across the section, the intensity of resistance per unit of area is called the intensely of stress or unit stress. In general usage the word stress is used to mean the intensity of stress.

Intensity of stress $\sigma = \dfrac{F}{A}$ …..(1.1)

σ = Stress N/m^2

A = Cross Section area in m^2

F = Total force acting through the centroidal axis of the body in N.

1.3 Strain: Strain is defined as the ratio of deformation of a body due to external load to the original length. It is a dimensionless parameter. If a load P is applied to a body the body deforms by δ and the original length be L the strain ($\in$) is given by

$$\in = \frac{\delta}{L} \qquad \text{.....(1.2)}$$

<u>1.4 Simple Stress & Strain</u>:

(a) Tensile Stress (σ_t): When a body is subjected to two equal and opposite pulls and the length of the body increases, the stress induced is called tensile stress (Fig 1.2).

The stress in (σ_t) simple tension in a body is equal to the external force P divided by the resisting area and in this case the resisting area is cross-sectional area A of the member in m^2. Thus

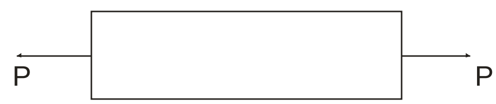


Fig. 1.2

$$\sigma_t = \frac{P}{A} \text{ N/m}^2 \qquad \text{.....(1.3)}$$

Corresponding strain is called tensile strain

$$\text{Tensile strain} = \frac{\text{Increase in length}}{\text{Original length}}$$

$$= e = \frac{dl}{l} \qquad \text{.....(1.4)}$$

(b) Compressive Stress (σ_C): When a body is subjected to two equal and opposite pushes and length of the body reduces, the stress induced is called compressive stress. (Fig 1.3). For a straight short compression member in which the resultant load P acts through centre of gravity of every cross section the stress σ_C

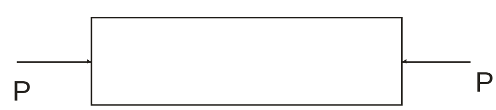


Fig. 1.3

$$\sigma_c = \frac{P}{A} \qquad \text{.....(1.5)}$$

Corresponding strain is called compressive strain.

$$\text{Compressive strain} = \frac{\text{Decrease in length}}{\text{Original length}}$$

$$e = \frac{dl}{l} \qquad \text{.....(1.6)}$$

(C) Shear Stress (τ): When equal and opposite forces are applied tangentially to any cross sectional plane of a body, tending to slide its one part over other, the body is said to be in state of shear. The corresponding stress induced is shear stress. In Fig.1.4 riveted joint tends to shear off at section XX due to externally applied load P. Shear stress is produced at the cross section of rivet. If the cross section area of rivet be A shear stress

$$\tau = \frac{P}{A} \qquad \text{.....(1.7)}$$

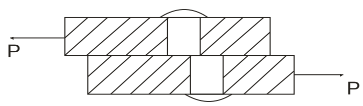
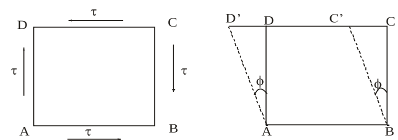


Fig. 1.4



(a) Before distortion (b) After distortion

Fig. 1.5

Shear Strain: It is a measure of the angle through which a body is deformed by the applied load. Let a rectangular block ABCD, fixed at bottom is distorted through an angle ϕ by the externally applied tangential load P to face CD as in Fig. 1.5. The face CD moves to new position C'D'. Let the movement of end C is very small.

Then shear strain = $\frac{CC'}{CB} = \tan\phi = \phi\, radian$ $\qquad$ [$\because \phi$ small]

<u>1.5 Elasticity</u>: A material is composed of small particles and molecules, between which forces are acting. These molecular forces resist the change in the shape of the body which external forces tend to produce. Under the action of external forces the particles of body are displaced and the displacement continues till the equilibrium is established between the external and internal forces.

As in Fig. 1.6 a bar is loaded at the end. Under the action of this load a certain elongation of the bar will take place. When the load is diminished the elongation of the bar diminishes and the loaded end of the bar moves upward. The property by which a body returns to its original shape after removal of the load is called ***elasticity***. The body is *perfectly elastic* if it recovers its original shape completely after unloading, *partially elastic* if the deformation disappear partially after unloading and *plastic* if the deformation is permanent.

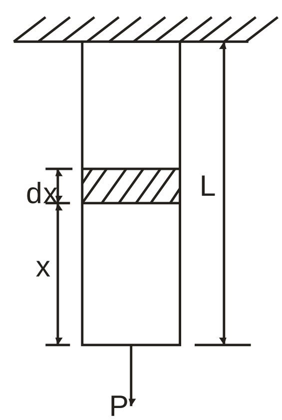


Fig. 1.6

1.6 Hooke's Law

It states that within an elastic limit stress is proportional to strain.

$$\text{Stress} \propto \text{Strain}$$

or $$\frac{\text{Stress}}{\text{Strain}} = E \qquad [\text{ C is a constant }]$$

'E' is known as coefficient of elasticity or modulus of elasticity. The ratio of tensile stress and tensile strain or compressive stress and compressive strain is called young's modulus of elasticity and is denoted by-.

$$\frac{\sigma_t}{\in_t} \quad or \quad \frac{\sigma_C}{\in_C} = E \qquad \text{.....(1.8)}$$

The ratio of shear stress and shear strain is called shear modulus or modulus of rigidity. It is denoted by C or G.

$$\frac{\tau}{\in_S} = C \qquad \text{.....(1.9)}$$

Let us consider a body subjected to a tensile force P, having length *l* and cross section area A, the stress induced

$$\sigma = \frac{P}{A}$$

And if the δ is the deformation and strain

$$\in_t = \frac{\delta}{l}$$

As per Hooke's law $$\frac{\sigma}{\in_t} = E$$

$$\frac{Pl}{A\delta} = E$$

or deformation $$\delta = \frac{Pl}{AE} \qquad \text{.....(1.10)}$$